

Examen de Habilidades 1: Redes enrutadad

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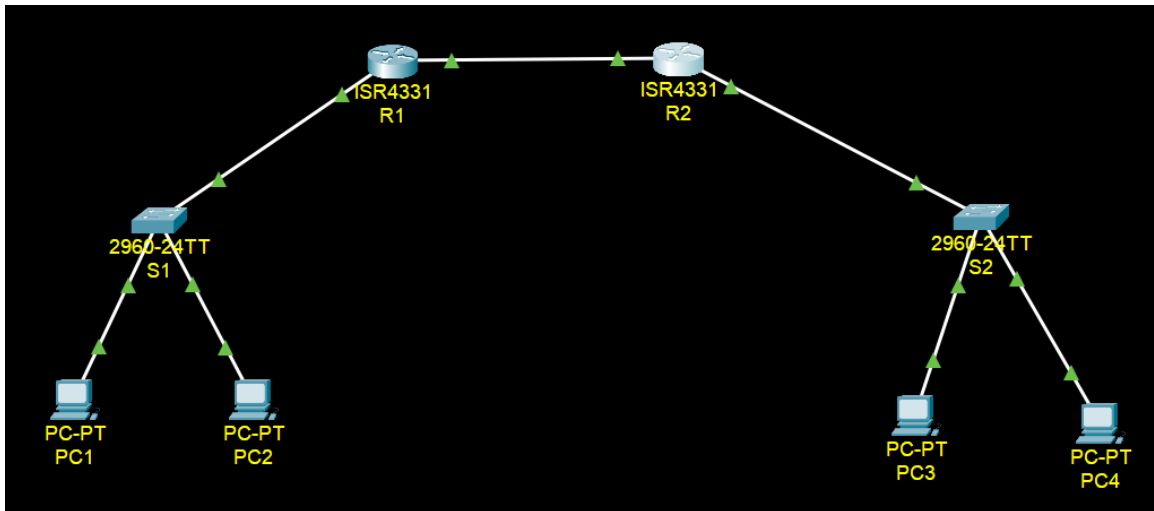


Figura 1. Topología a crear.

Objetivos

- Crear la red y configurar los parámetros básicos de los dispositivos.
- Configurar y verificar OSPFv2 de área única para el funcionamiento del enrutamiento.
- Optimizar y verificar la configuración OSPFv2 de área única.

Información básica/situación

Se debe utilizar los rúteres sin interfaces seriales.

Los suiches indicados en la topología deben tener configurada la SVI.

Se requiere configurar la red de una pequeña empresa mediante OSPFv2.

Recursos necesarios

- Computador personal con Windows
- Conexión a Internet

Procedimiento

Parte 1: Crear la red enrutada

Parte 2: Realice la configuración inicial de los dispositivos intermediarios

Parte 3: Configuración del enrutamiento OSPF

Cuestionario

Con respecto al procedimiento realizado, responda las siguientes preguntas:

1. Defina el esquema de direcciones para las redes.

Dipositivo	Ip	Masacara	gateway
Pc0	192.168.31.10	255.255.255.0	192.168.31.1
Pc1	192.168.31.11	255.255.255.0	192.168.31.1
Pc2	192.168.33.10	255.255.255.0	192.168.33.1
Pc3	192.168.33.11	255.255.255.0	192.168.33.1
Router1	192.168.32.1	255.255.255.0	192.168.32.1
Router2	192.168.33.1	255.255.255.0	192.168.32.2
Switch1	192.168.31.2	255.255.255.0	192.168.31.1
Switch2	192.168.33.2	255.255.255.0	192.168.33.1

2. Incluya la configuración inicial de los rúteres.

Examen de Habilidades 2: Creación de una red local

 Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: n

Press RETURN to get started!

Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname rl
rl(config)#no ip domain
rl(config)#no ip domain
rl(config)#no ip domain-lo
rl(config)#no ip domain-lookup
rl(config)#interface g0/0/0
rl(config-if)#ip address 192.168.31.1 255.255.255.0
rl(config-if)#no shuw
rl(config-if)#no shut
rl(config-if)#no shutdown

rl(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to u

rl(config-if)#
rl(config-if)#interface g0/1
%Invalid interface type and number
rl(config)#interface g0/0/1
rl(config-if)#ip address 192.168.32.1 255.255.255.252
rl(config-if)#no shuw
rl(config-if)#no shut
rl(config-if)#no shutdown

rl(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up
```

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Examen de Habilidades 2: Creación de una red local

```
Router1
Physical Config CLI Attributes
IOS Command Line Interface
Would you like to enter the initial configuration dialog? [yes/no]:
% Please answer 'yes' or 'no'.
Would you like to enter the initial configuration dialog? [yes/no]: no
Press RETURN to get started!
Router>enable
Router#configure t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#hostname r2
r2(config)#no ip domain
r2(config)#no ip domain-
r2(config)#no ip domain-lo
r2(config)#no ip domain-lookup
r2(config)#interface g0/0/0
r2(config-if)#ip address 192.168.33.1 255.255.255.0
r2(config-if)#no shut
r2(config-if)#no shutdown
r2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
r2(config-if)#
r2(config-if)#interface
r2(config-if)#interface g0/0/1
r2(config-if)#ip add
r2(config-if)#ip address 192.168.32.2 255.255.255.252
r2(config-if)#no
r2(config-if)#no shu
r2(config-if)#no shutdown
r2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
```

3. Incluya la configuración inicial de los suiches.

Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Switch>
Switch>
Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#hostname S1
S1(config)#
S1(config)#
S1(config)#interface vlan1
S1(config-if)#ip address 192.168.31.2 255.255.255.0
S1(config-if)#no shu
S1(config-if)#no shutdown

S1(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

S1(config-if)#
S1(config-if)#ip default
S1(config-if)#ip default-
S1(config-if)#ip default-gate
S1(config-if)#ip dafault-gateway 192.168.31.1
      ^
% Invalid input detected at '^' marker.

S1(config-if)#exit
S1(config)#
S1(config)#ip default-
S1(config)#ip default-gateway 192.168.31.1
S1(config)#
```

Examen de Habilidades 2: Creación de una red local

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

Switch	Ports	Model	SW Version	SW Image
*	1 26	WS-C2960-24TT-L	15.0(2)SE4	C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: <http://www.cisco.com/techsupport>
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

Switch>
Switch>
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname S2
S2(config)#interface vlan2
S2(config-if)#ip address 192.168.33.2 255.255.255.0
S2(config-if)#no shu
S2(config-if)#no shutdown
S2(config-if)#
S2(config-if)#exit
S2(config)#ip default
S2(config)#ip default-gateway 192.168.33.1
S2(config)#

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4. Incluya la configuración de la PC1 y la PC4.

Examen de Habilidades 2: Creación de una red local

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.31.10

Subnet Mask 255.255.255.0

Default Gateway 192.168.31.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::20D:BDFF:FE5C:D198

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Examen de Habilidades 2: Creación de una red local

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.31.11

Subnet Mask 255.255.255.0

Default Gateway 192.168.31.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::260:70FF:FECA:AC1B

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Examen de Habilidades 2: Creación de una red local

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.33.10

Subnet Mask 255.255.255.0

Default Gateway 192.168.33.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:42FF:FE84:C482

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Examen de Habilidades 2: Creación de una red local

The screenshot shows the 'PC3' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is active, showing settings for the 'FastEthernet0' interface. The 'Static' radio button is selected for both IPv4 and IPv6 configurations. The IPv4 configuration includes an IP address of 192.168.33.11, a subnet mask of 255.255.255.0, a default gateway of 192.168.33.1, and a DNS server of 0.0.0.0. The IPv6 configuration includes a static address, a link local address of FE80::2D0:BAFF:FE22:724E, and empty fields for default gateway and DNS server. The 802.1X section shows 'Use 802.1X Security' is unchecked, with 'Authentication' set to 'MD5' and empty fields for 'Username' and 'Password'.

PC3

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.33.11

Subnet Mask 255.255.255.0

Default Gateway 192.168.33.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2D0:BAFF:FE22:724E

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

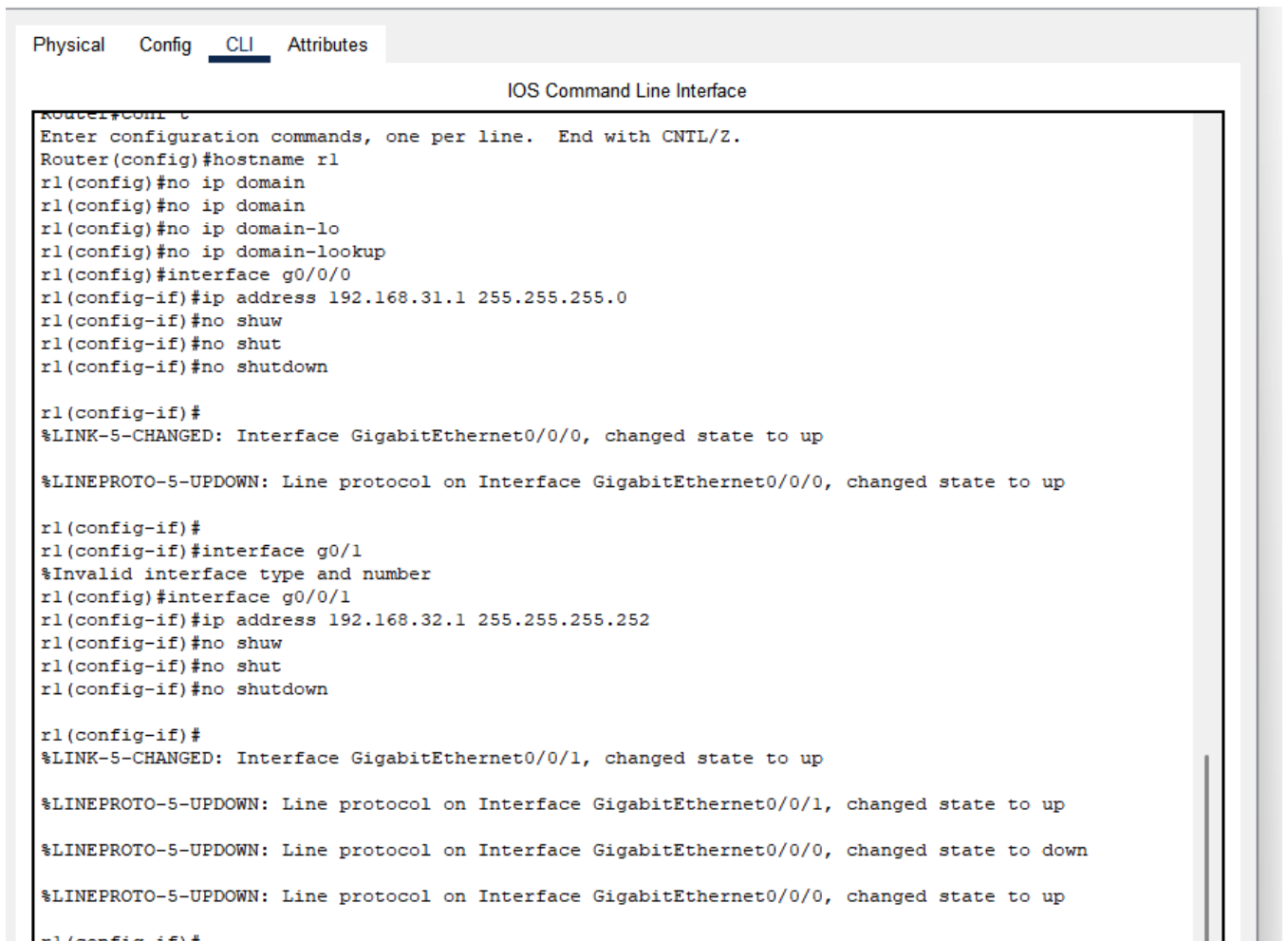
Authentication MD5

Username

Password

5. Incluya la configuración de las interfaces del router R1.

Examen de Habilidades 2: Creación de una red local

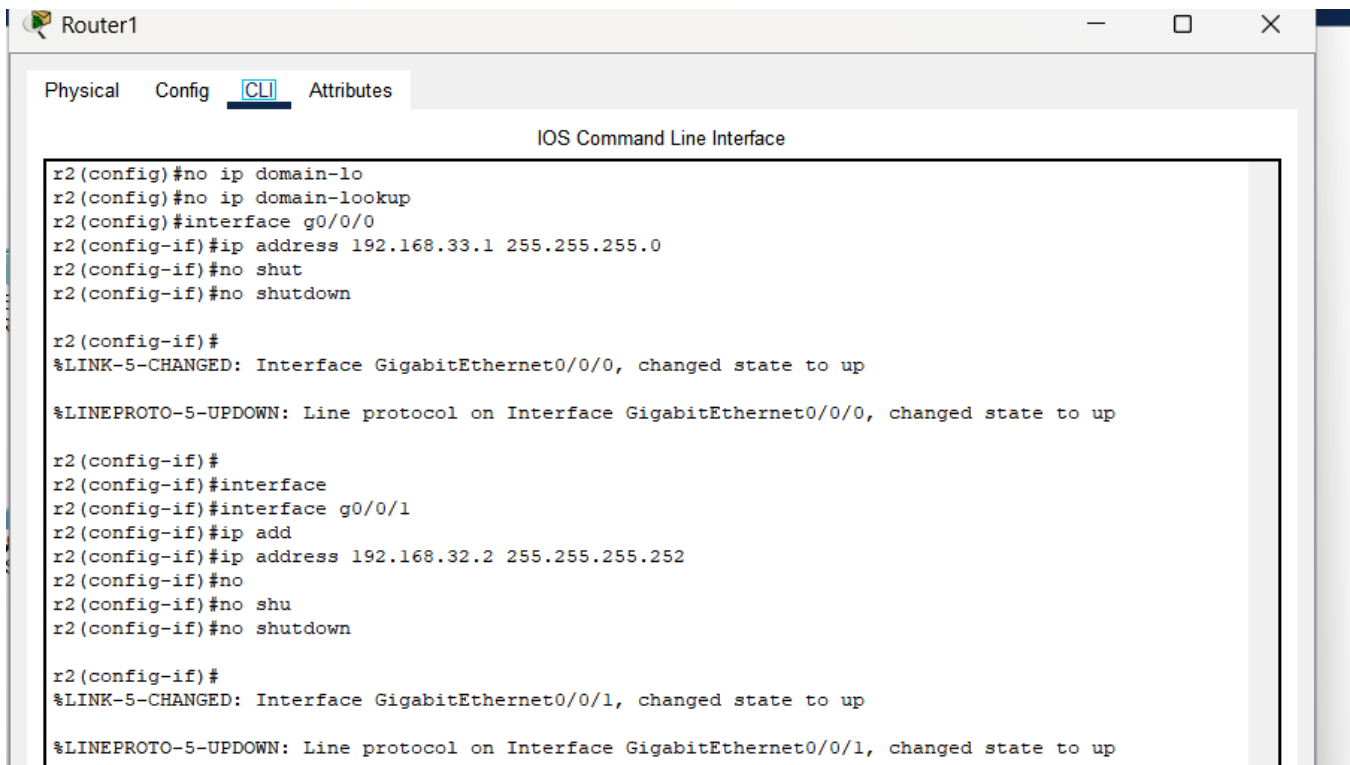


The screenshot displays the Cisco IOS Command Line Interface (CLI) for a router. The tabs at the top are 'Physical', 'Config', 'CLI' (selected), and 'Attributes'. The title bar reads 'IOS Command Line Interface'. The command prompt is 'Router#'. The user enters 'conf t' to enter configuration mode, resulting in 'Router(config)#'. The user then enters 'hostname r1', resulting in 'r1(config)#'. The user enters 'no ip domain', resulting in 'r1(config)#no ip domain'. The user enters 'no ip domain-lookup', resulting in 'r1(config)#no ip domain-lookup'. The user enters 'interface g0/0/0', resulting in 'r1(config-if)#'. The user enters 'ip address 192.168.31.1 255.255.255.0', resulting in 'r1(config-if)#ip address 192.168.31.1 255.255.255.0'. The user enters 'no shu', resulting in 'r1(config-if)#no shu'. The user enters 'no shutdown', resulting in 'r1(config-if)#no shutdown'. The user enters '#', resulting in '%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up'. The user enters '#', resulting in '%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up'. The user enters '#', resulting in 'r1(config-if)#'. The user enters 'interface g0/1', resulting in '%Invalid interface type and number'. The user enters 'interface g0/0/1', resulting in 'r1(config-if)#'. The user enters 'ip address 192.168.32.1 255.255.255.252', resulting in 'r1(config-if)#ip address 192.168.32.1 255.255.255.252'. The user enters 'no shu', resulting in 'r1(config-if)#no shu'. The user enters 'no shut', resulting in 'r1(config-if)#no shut'. The user enters 'no shutdown', resulting in 'r1(config-if)#no shutdown'. The user enters '#', resulting in '%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up'. The user enters '#', resulting in '%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up'. The user enters '#', resulting in '%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to down'. The user enters '#', resulting in '%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up'. The user enters '#', resulting in 'r1(config-if)#'.

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname r1
r1(config)#no ip domain
r1(config)#no ip domain-lookup
r1(config)#no ip domain-lookup
r1(config)#interface g0/0/0
r1(config-if)#ip address 192.168.31.1 255.255.255.0
r1(config-if)#no shu
r1(config-if)#no shutdown
r1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
r1(config-if)#
r1(config-if)#interface g0/1
%Invalid interface type and number
r1(config)#interface g0/0/1
r1(config-if)#ip address 192.168.32.1 255.255.255.252
r1(config-if)#no shu
r1(config-if)#no shut
r1(config-if)#no shutdown
r1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
r1(config-if)#
```

6. Incluya la configuración de las interfaces del router R2.

Examen de Habilidades 2: Creación de una red local



```
r2(config)#no ip domain-lo
r2(config)#no ip domain-lookup
r2(config)#interface g0/0/0
r2(config-if)#ip address 192.168.33.1 255.255.255.0
r2(config-if)#no shut
r2(config-if)#no shutdown

r2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

r2(config-if)#
r2(config-if)#interface
r2(config-if)#interface g0/0/1
r2(config-if)#ip add
r2(config-if)#ip address 192.168.32.2 255.255.255.252
r2(config-if)#no
r2(config-if)#no shu
r2(config-if)#no shutdown

r2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
```

7. Incluya la configuración de enrutamiento OSPF de R1.

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

r1(config-if)#
r1(config-if)#
r1(config-if)#
r1(config-if)#exit
r1(config)#router ospf 1
r1(config-router)#router 1.1.1.1
r1(config-router)#new
r1(config-router)#netw
r1(config-router)#network 192.168.31.0 0.0.0.255 area 0
r1(config-router)#network 192.168.32.0 0.0.0.3 area 0
r1(config-router)#
```

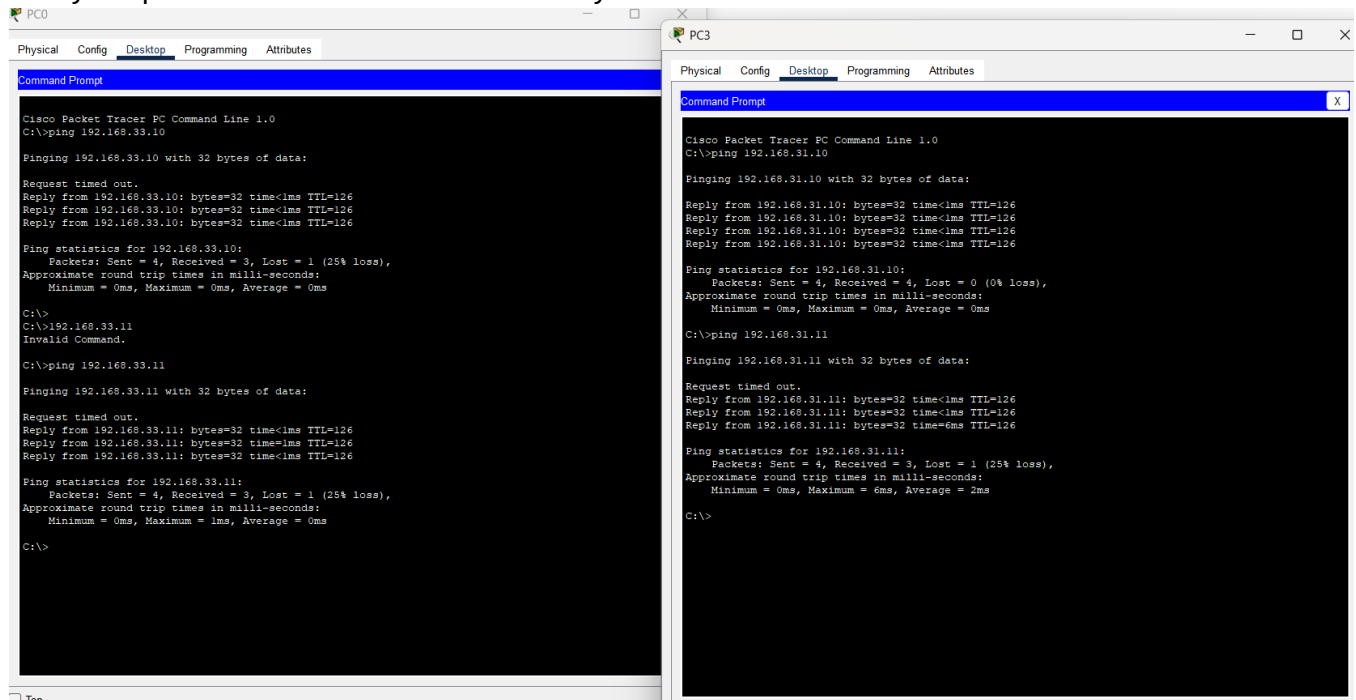
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8. Incluya la configuración de enrutamiento OSPF de R2.

Examen de Habilidades 2: Creación de una red local

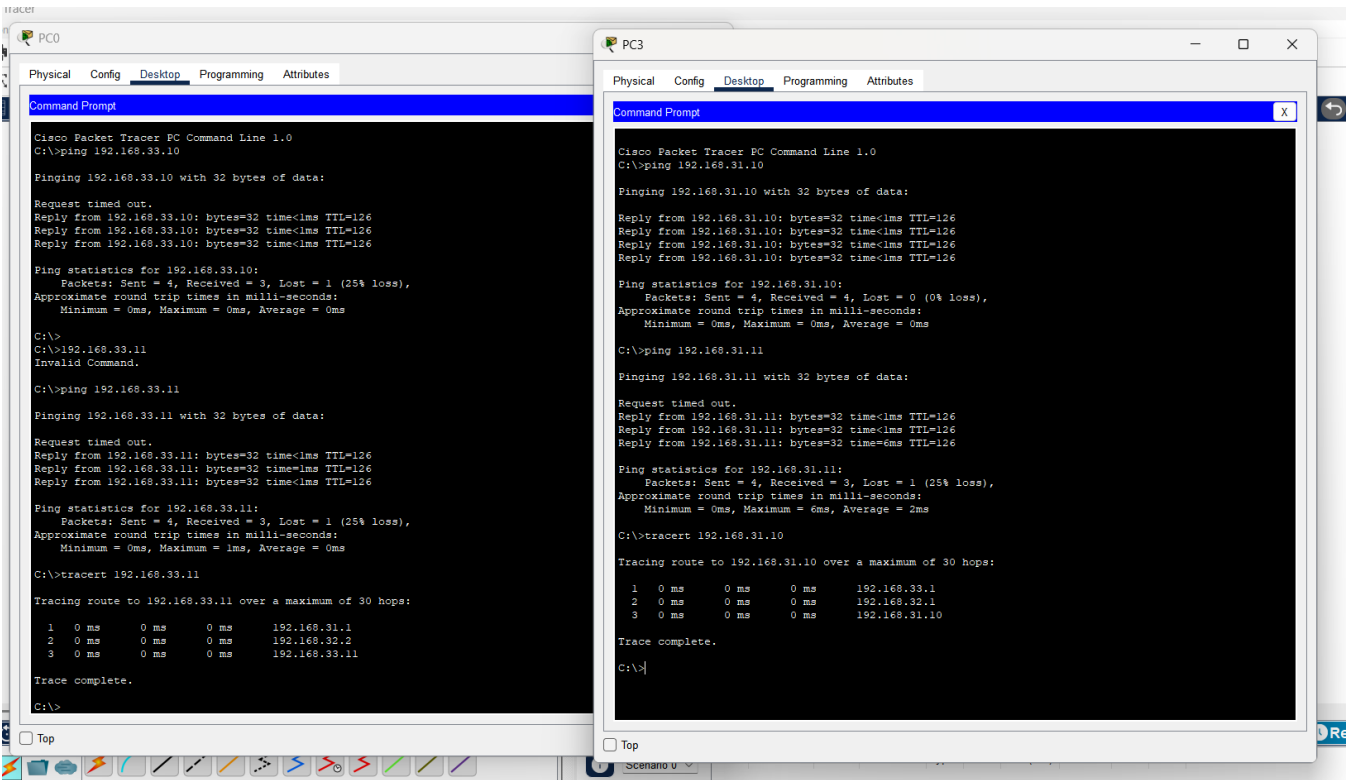
```
Invalid input detected at '^' marker.  
  
r2(config)#router ospf 1  
r2(config-router)#router-id 2.2.2.2  
r2(config-router)#net  
r2(config-router)#network 192.168.33.0 0.0.0.255 area 0  
r2(config-router)#network 192.168.32.0 0.0.0.3 area 0  
r2(config-router)#
```

9. Incluya la prueba de conectividad entre PC1 y PC4.



10. Incluya el trazado de ruta entre PC1 y PC4.

Examen de Habilidades 2: Creación de una red local



Calificación

Para la calificación se tomará en cuenta la documentación del procedimiento con capturas de pantalla que evidencien lo siguiente:

Criterio		Puntaje
1	Creación de la red enrutada	2
2	Configuración de los dispositivos intermediarios	2
3	Configuración de las interfaces de los rúteres.	2
4	Configuración de enrutamiento OSPF	2
5	Prueba de conectividad entre PC1 Y PC4	12
TOTAL		20

